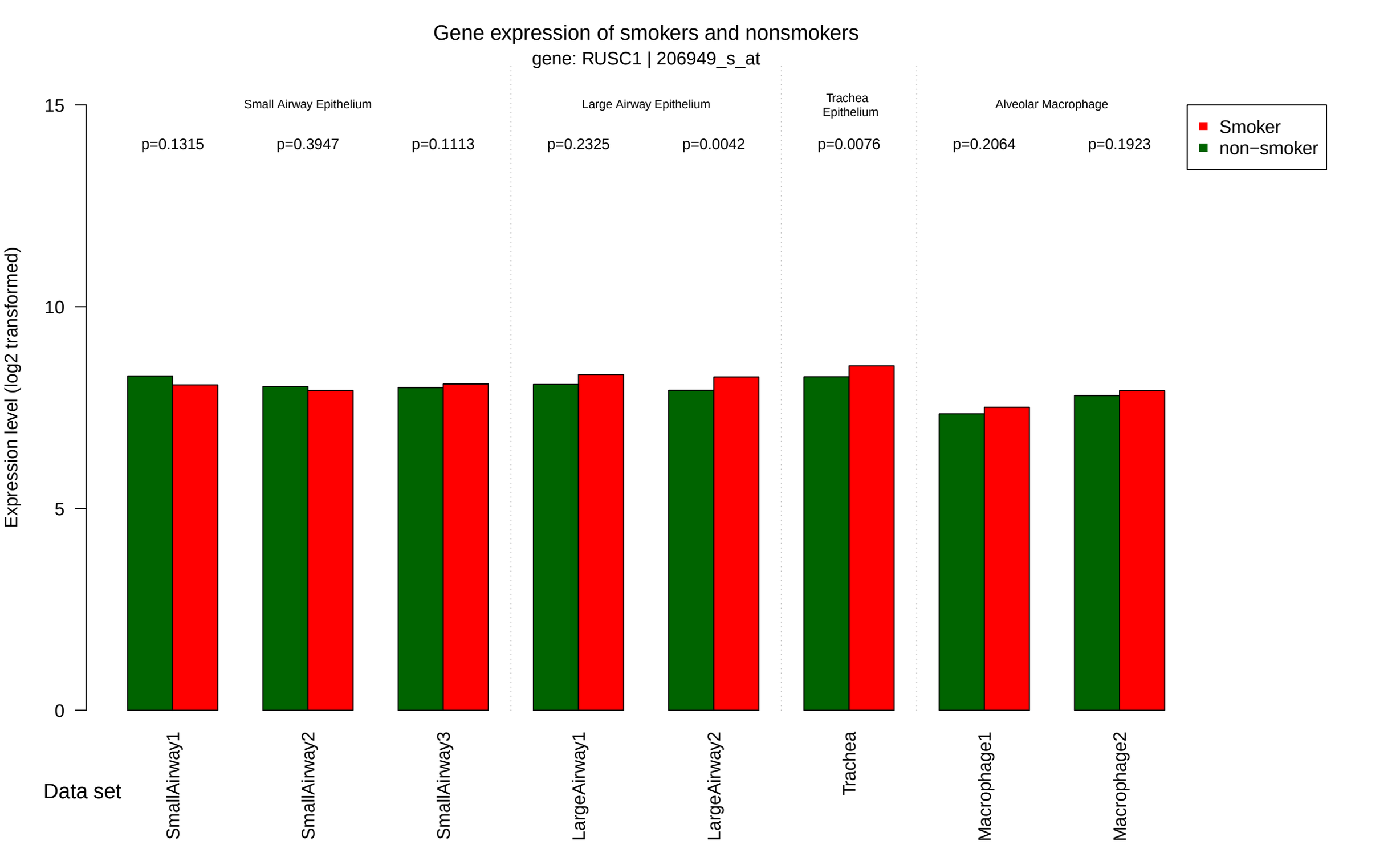
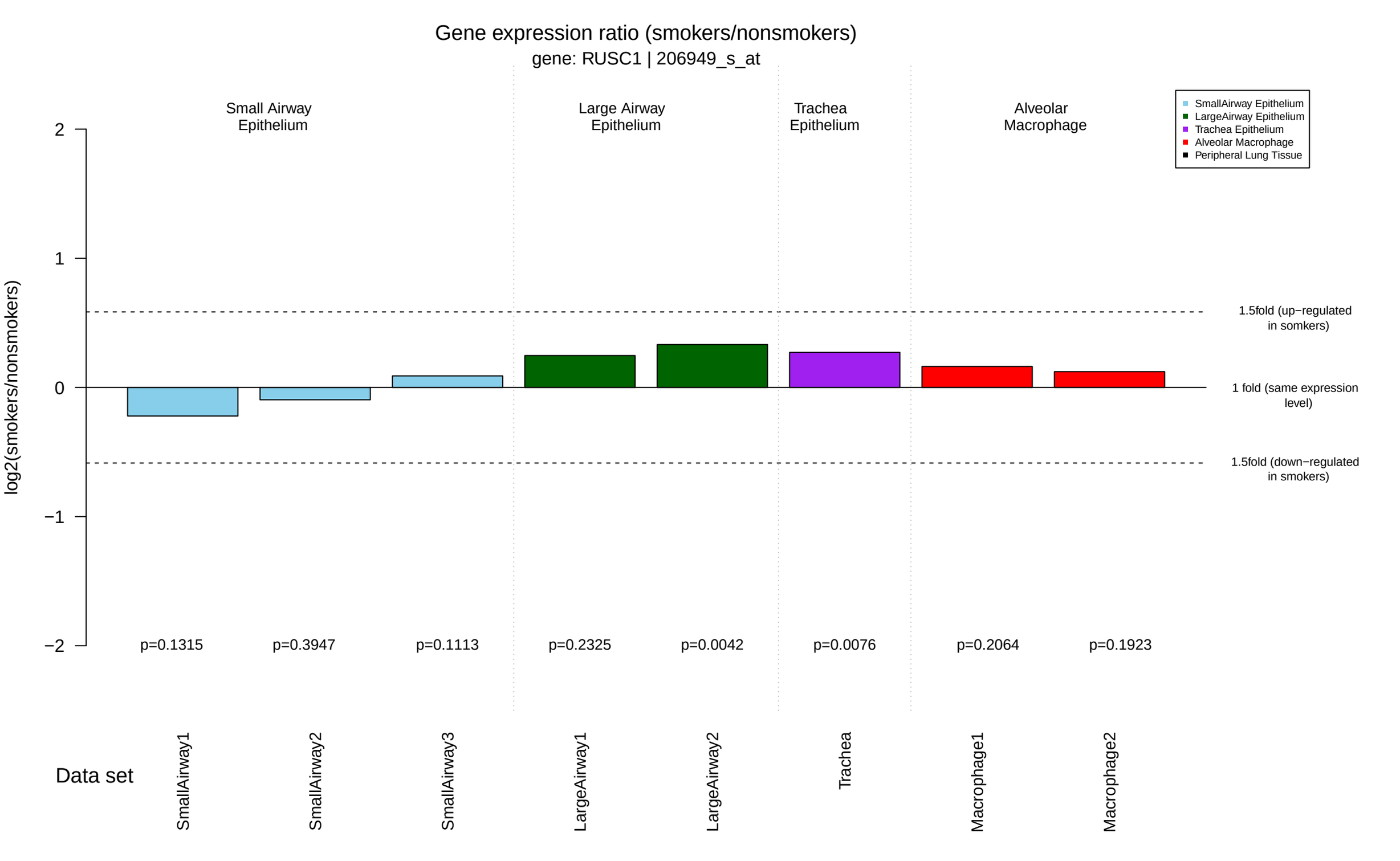
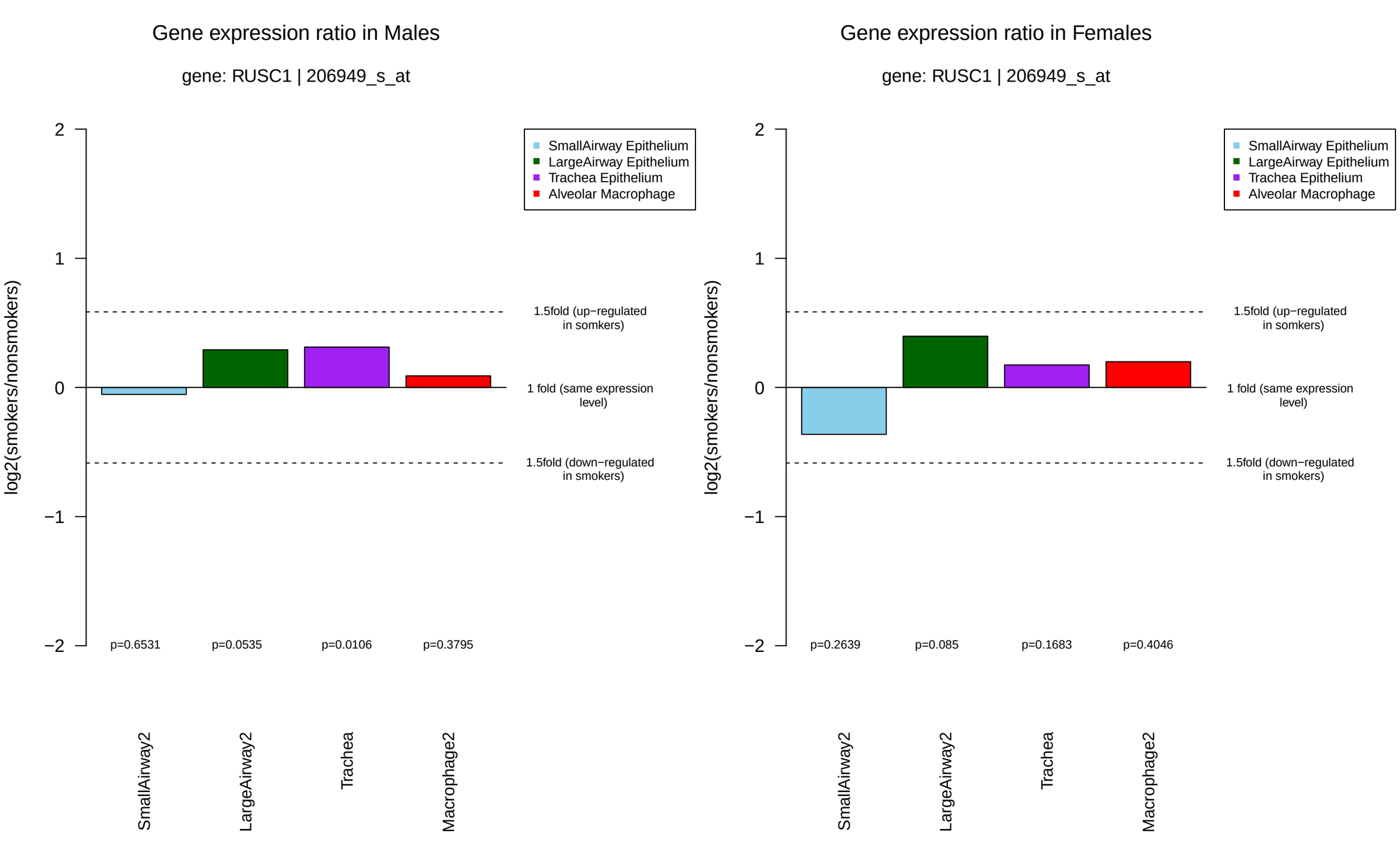
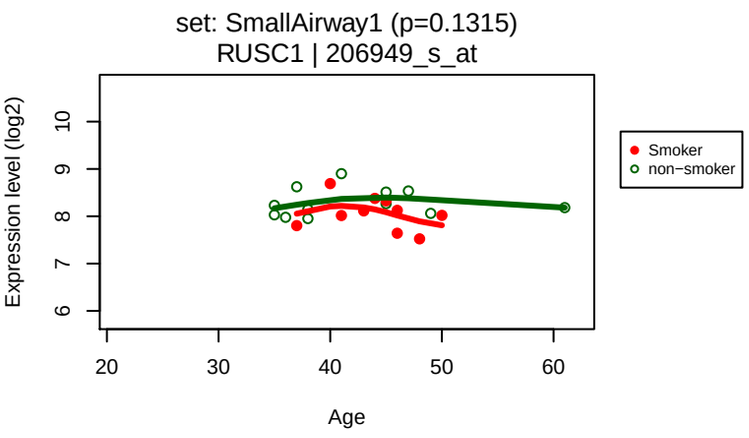
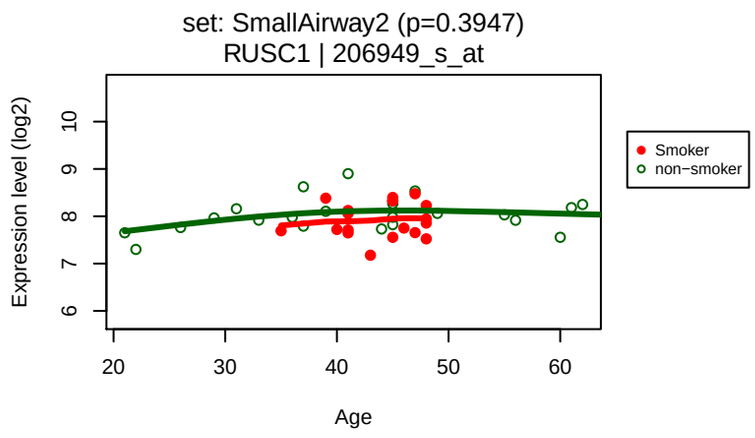
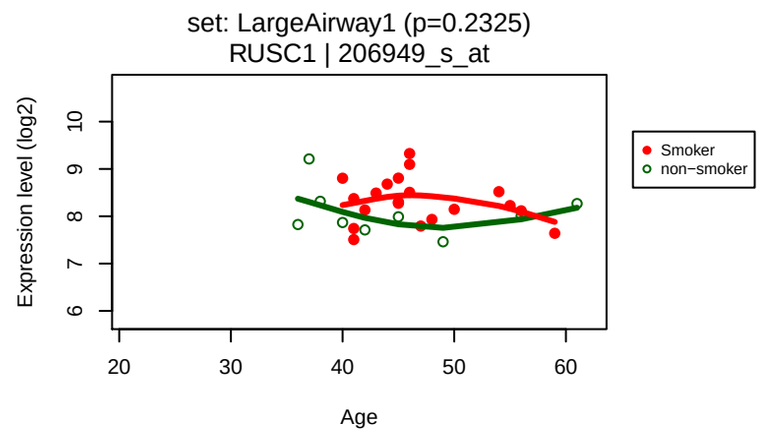

Gene expression of smokers and nonsmokers
gene: RUSC1 | 206949_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1315
p=0.3947
p=0.1113
p=0.2325
p=0.0042
p=0.0076
p=0.2064
p=0.1923
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

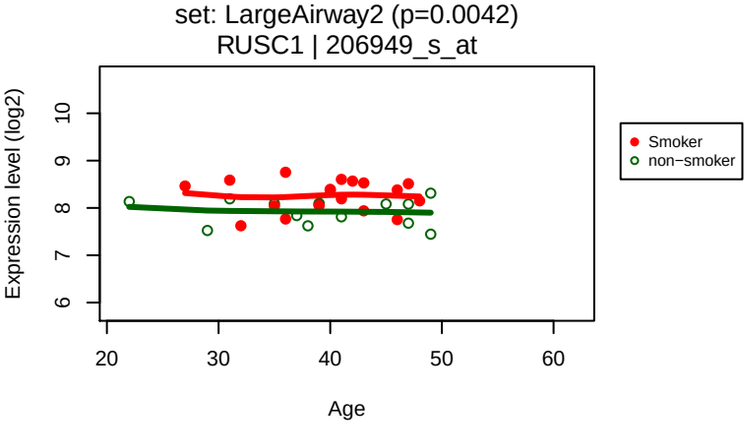

Gene expression ratio (smokers/nonsmokers)
gene: RUSC1 | 206949_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1315
p=0.3947
p=0.1113
p=0.2325
p=0.0042
p=0.0076
p=0.2064
p=0.1923
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

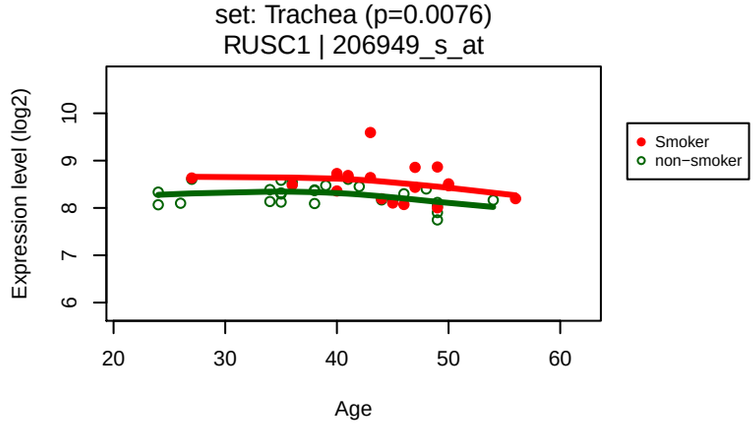

Gene expression ratio in Males
gene: RUSC1 | 206949_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6531
p=0.0535
p=0.0106
p=0.3795
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUSC1 | 206949_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2639
p=0.085
p=0.1683
p=0.4046
SmallAirway2
LargeAirway2
Trachea
Macrophage2

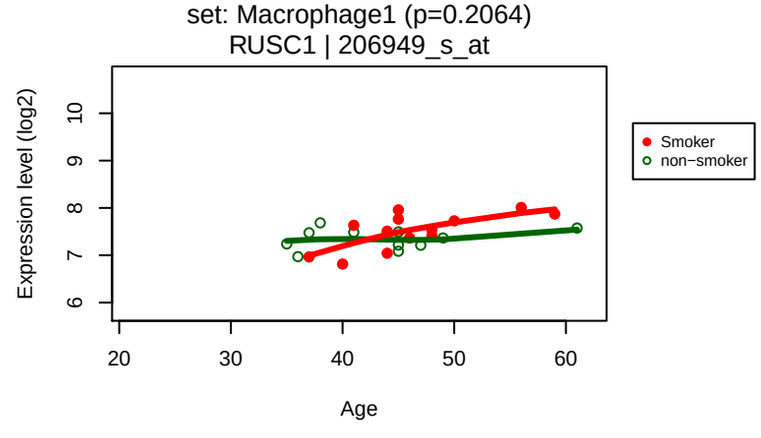

set: SmallAirway1 (p=0.1315)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker

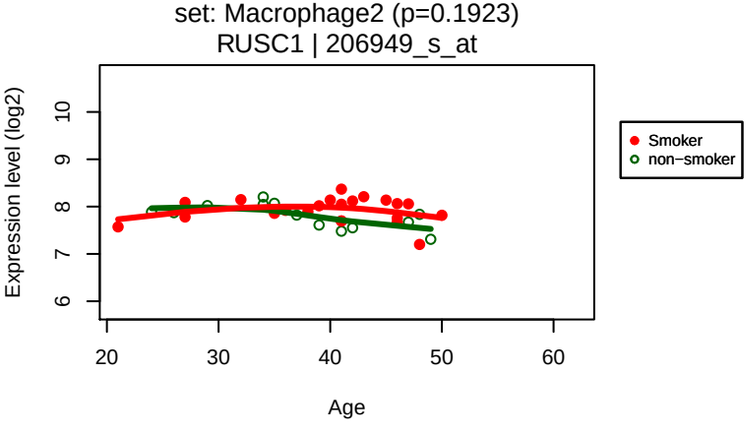

set: SmallAirway2 (p=0.3947)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker

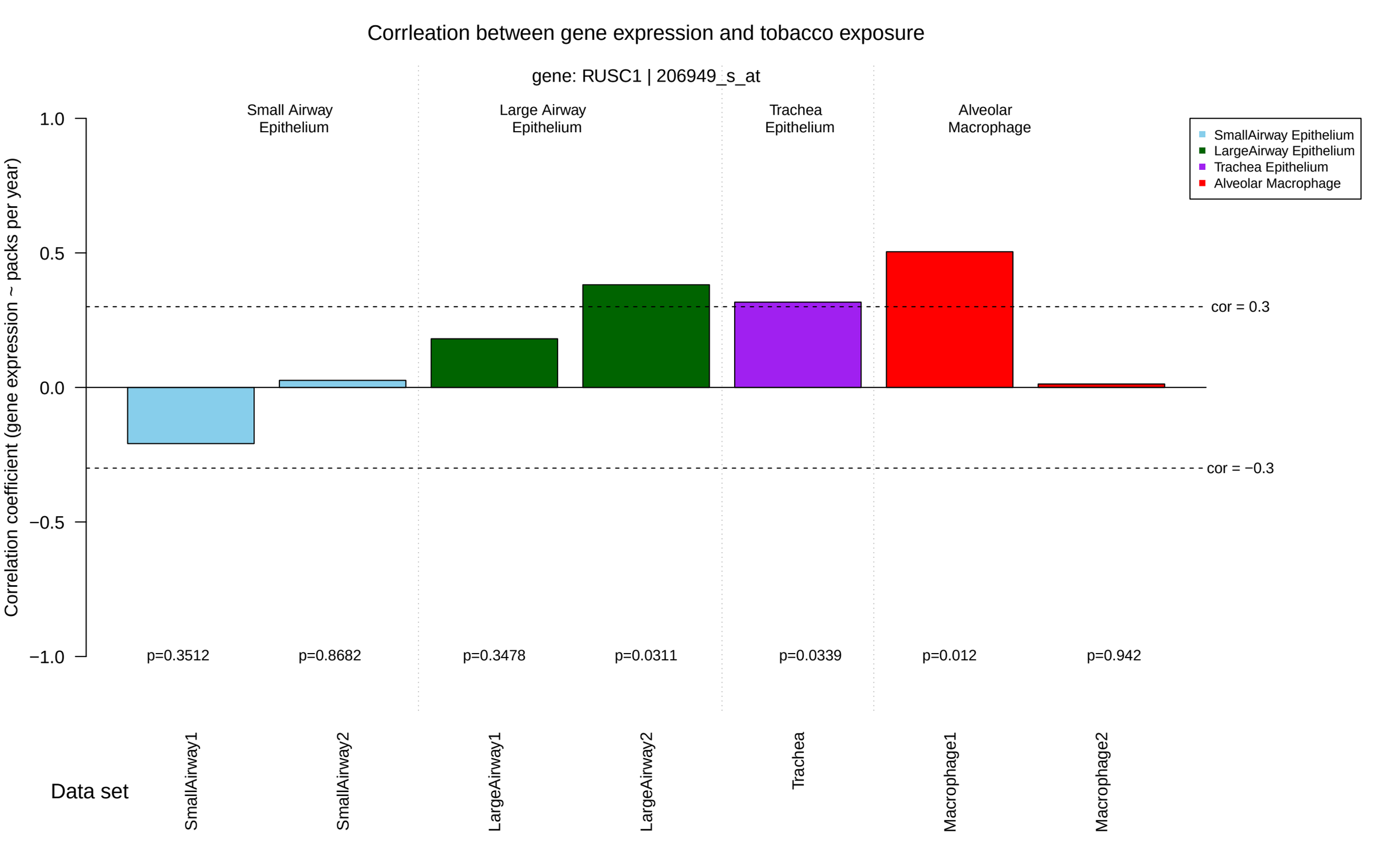

set: LargeAirway1 (p=0.2325)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0042)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0076)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2064)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1923)
RUSC1 | 206949_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUSC1 | 206949_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3512
p=0.8682
p=0.3478
p=0.0311
p=0.0339
p=0.012
p=0.942
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2